

Delta Lake Incremental Processing & SCD Assignment Summary

Implementation details for incremental data loading, SCD Type 1 & Type 2 MERGE operations, and validation metrics.

1. Data Pipeline Overview

This project processes customer records incrementally using Delta Lake. Base records are cleaned and loaded from customer_master.csv. Incremental updates from customer_incremental.csv are merged using SCD Type 1 and SCD Type 2 logic.

2. Validation & Record Counts

Dataset / Stage	Record Count	Key Uniqueness	Status
customer_master.csv	793	Unique Primary Keys	Cleaned
customer_incremental.csv (Raw)	50	Includes 5 Duplicates	Sanitized to 45
SCD Type 1 Post-Merge Active	808	0 Duplicate Keys	Verified
SCD Type 2 Total Table Rows	810	1 History + 808 Active	Verified (100%)

3. Core Implementations

- **SCD Type 1:** Overwrites updated attributes (address, sales, credit score) for existing matched customer keys and inserts new customer keys.
- **SCD Type 2:** Closes historical active record (Is_Active = False, End_Date = '2026-08-02') and appends new active version (Is_Active = True, Record_Version = 3).
- **DuckDB Analytics:** Queries active table state directly via delta_scan() for fast aggregated metrics.